

JAVA Programming Assignment - 3

(Submitted by Chaitanya Jindal)

Q1. Create a small application in Java for a College Management System.

- Design a class named Teacher with the following attributes:
 - Name
 - Qualification
- Extend the Teacher class into a derived class named Department, which contains:
 - Department Number
 - Department Name
- Create an interface named College with the following:
 - A field representing the name of the college
 - Two abstract methods: `getDetails()` and `showDetails()`
 - One default method to display a general message
 - One static method to display a general message
- Override the base class and interface methods in the derived class (Department).
- Ask the user to enter the department name and validate it:
 - It must be either "IT dept" or "Management dept"
 - If it does not match, display: "Not a valid Record"
- Assign department numbers based on input:
 - IT dept → Department Number = 10
 - Management dept → Department Number = 20
- Using the above classes and interface, collect and display the appropriate information.
- Create a class named CMS containing the main method, where all functionalities are executed.

```
package Assignment3;
import java.util.*;
interface College {
    String CollegeName = "CDAC";
    void getDetails();
    void showDetails();
    default void generalMessage() {
        System.out.println("Welcome to CMS");
    }
    static void displayRules() {
        System.out.println("Please enter valid department details...");
    }
}
class Teacher {
    String name = null;
    String qualification = null;
```

```
(base) PS D:\Projects\CDAC-PGCP-AI\Java Programming> javac Assignment3\CMS.java
(base) PS D:\Projects\CDAC-PGCP-AI\Java Programming> java Assignment3\CMS
Welcome to CMS
Please enter valid department details...
Enter Teacher Name: ABC
Enter the qualification: Ph.D
Enter Department Name: Management
Details:-
College Name: CDAC
Teacher Name: ABC
Department Name: Management
Department Number: 20.0
```



```

void teacherDetails(String name, String qualification) {
    this.name = name;
    this.qualification = qualification;
}
}
class Department extends Teacher implements College {
    double deptno;
    String deptname = null;
    Scanner sc = new Scanner(System.in);
    @Override
    public void getDetails() {
        System.out.print("Enter Teacher Name: ");
        name = sc.next();
        System.out.print("Enter the qualification: ");
        qualification = sc.next();
        System.out.print("Enter Department Name: ");
        deptname = sc.next();
        if (deptname.equalsIgnoreCase("IT")) {
            deptno = 10;
        } else if (deptname.equalsIgnoreCase("Management")) {
            deptno = 20;
        } else {
            System.out.println("Not a Valid Record.");
            System.exit(0);}
    }
    @Override
    public void showDetails() {
        System.out.println("Details:-");
        System.out.println("College Name: " + CollegeName);
        System.out.println("Teacher Name: " + name);
        System.out.println("Department Name: " + deptname);
        System.out.println("Department Number: " + deptno);
    }
}
public class CMS {
    public static void main(String[] args) {
        Department d = new Department();
        d.generalMessage();
        College.displayRules();
        d.getDetails();
        d.showDetails();}
}

```

